Hilbert Symbol

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Abstract

This note refers to Chapter III of GTM7, A Course in Arithmetic by J.-P. Serre, and the notes Introduction to the Local-Global Principle by Liang Xiao.

I presented Hilbert symbol in the Algebra & Number Theory seminar, organized by Liang Xiao in Peking University on May 10th, 2026.

Contents

1 Preliminaries

Let q be a power of a prime number p, and K be a field with q elements.

Theorem 1.1 (Chevalley-Warning)**.** *Let $f_\alpha \in K[X_1, \cdots, X_n]$ be polynomials such that $\sum_\alpha \deg f_\alpha < n$, and let V be the set of their common zeros in K^n. Then*

$$|V| \equiv 0 \pmod p.$$

Proof. $P = \prod_\alpha (1 - f_\alpha^{q-1})$ and $x \in K^n$. Then P is the characteristic function of V and denote $S(f) = \sum_{x \in K^n} f(x)$. We have

$$|V| \equiv S(P) \pmod p.$$

Since $\deg P < n(q-1)$, P is a linear combination of $X^u = X_1^{u_1} \cdots X_n^{u_n}$ with $\sum u_i < n(q-1)$. For every monomial X^u, we have $S(X^u) = 0$ since at least one $u_i < q-1$. □

Corollary 1.2. *If $\sum_\alpha \deg f_\alpha < n$ and f_α have no constant term, then f_α have a nontrivial common zero.*

Proof. 0 is a solution. □

Corollary 1.3. *All quadratic forms in at least 3 variables over K have a nontrivial zero.*

Definition 1.4. Let $p \neq 2$ be a prime number, and let $x \in \mathbb{F}_p^*$. The **Legendre symbol** of x, denoted by $\left(\frac{x}{p}\right)$, is the integer $x^{\frac{p-1}{2}} = \pm 1$.

If n is an odd integer, let $\varepsilon(n)$ and $\omega(n)$ be the elements of $\mathbb{Z}/2\mathbb{Z}$ defined by

$$\varepsilon(n) \equiv \frac{n-1}{2} \mod 2 = \begin{cases} 0 & \text{if } n \equiv 1 \pmod 4, \\ 1 & \text{if } n \equiv -1 \pmod 4, \end{cases}$$

$$\omega(n) \equiv \frac{n^2-1}{8} \pmod 2 = \begin{cases} 0 & \text{if } n \equiv \pm 1 \pmod 8, \\ 1 & \text{if } n \equiv \pm 5 \pmod 8. \end{cases}$$

Theorem 1.5 (Gauss's Quadratic Reciprocity Law)**.** *Let l, p be distinct odd primes different from 2.*

$$\left(\frac{l}{p}\right) = (-1)^{\varepsilon(p)\varepsilon(l)} \left(\frac{l}{p}\right).$$

1.1 The Ring $\mathbb{Z}_p$ and the Field $\mathbb{Q}_p$

Let $A_n = \mathbb{Z}/p^n\mathbb{Z}$, and $\phi_n : A_n \to A_{n-1}$ be the natural projection.

The ring of p-adic integers $\mathbb{Z}_p = \varprojlim (A_n, \phi_n)$ and we let $\varepsilon_n : \mathbb{Z}_p \to A_n$ be the natural projection.

The field of p-adic numbers $\mathbb{Q}_p$ is the field of fractions of $\mathbb{Z}_p$.

Proposition 1.6. *Let U denotes the group of invertible elements in $\mathbb{Z}_p$. Then every nonzero element of $\mathbb{Z}_p$ can be written uniquely as $p^n u$ with $n \geqslant 0$ and $u \in U$.*

1.2 p-adic Equations

Lemma 1.7. *Let* $\cdots \to D_n \to D_{n-1} \to \cdots \to D_1$ *be a projective system. If* D_n *are nonempty finite sets, then* $D = \varprojlim D_n$ *is nonempty.*

Proof. □

If $f \in \mathbb{Z}_p[X_1, \cdots, X_m]$ is a polynomial with p-adic integer coefficients, we denote by f_n the image of f in $A_n[X_1, \cdots, X_m]$.

Proposition 1.8. *Let* $f^{(i)} \in \mathbb{Z}_p[X_1, \cdots, X_m]$ *be polynomials with p-adic integer coefficients. The following are equivalent:*

1. *The* $f^{(i)}$ *have a common zero in* $(\mathbb{Z}_p)^m$.
2. *For every* $n > 1$*, the* $f_n^{(i)}$ *have a common zero in* $(A_n)^m$.

Proof. Let $D(D_n)$ be the set of common zeros of the $f^{(i)}$ $(f_n^{(i)})$ in $(\mathbb{Z}_p)^m$ $((A_n)^m)$. D_n are finite and $D = \varprojlim D_n$. By Lemma 1.7, $D \neq \varnothing \iff D_n \neq \varnothing$ for all n. □

Definition 1.9. A point $x = (x_1, \cdots, x_m) \in (\mathbb{Z}_p)^m$ is **primitive** if one of the x_i is a invertible, i.e. if x_i are not all divisible by p.

Proposition 1.10. *Let* $f^{(i)} \in \mathbb{Z}_p[X_1, \cdots, X_m]$ *be homogeneous polynomials with p-adic integer coefficients. The following are equivalent:*

1. *The* $f^{(i)}$ *have a nontrivial common zero in* $(\mathbb{Q}_p)^m$.
2. *The* $f^{(i)}$ *have a common primitive zero in* $(\mathbb{Z}_p)^m$.
3. *For every* $n > 1$*, the* $f_n^{(i)}$ *have a common primitive zero in* $(A_n)^m$.

Theorem 1.11. *Let* $f \in \mathbb{Z}_p[X_1, \cdots, X_m], x = (x_i) \in (\mathbb{Z}_p)^m, n, k, j \in \mathbb{Z}$ *and* $0 \leqslant j \leqslant m$*. Suppose that* $0 < 2k < n$ *and that*

$$f(x) \equiv 0 \pmod{p^n}, \quad v_p\left(\frac{\partial f}{\partial X_i}(x)\right) = k.$$

Then there exists a zero y *of* f *in* $(\mathbb{Z}_p)^m$ *such that* $y \equiv x \pmod{p^{n-k}}$.

Corollary 1.12. *Suppose* $p \neq 2$*. Let* $f(X) = \sum a_{ij} X_i X_j$ *with* $a_{ij} = a_{ji}$ *be a quadratic form with coefficient in* $\mathbb{Z}_p$ *with* $\det(a_{ij})$ *invertible. Let* $a \in \mathbb{Z}_p$*. Every primitive solution of the equation* $f(x) \equiv a \pmod{p}$ *lifts to a true solution.*

1.3 $\mathbb{Q}_p^*$

Let $U = \mathbb{Z}_p^*$ be the group of p-adic units and $U_n = 1 + p^n\mathbb{Z}_p$, which is the kernel of $\varepsilon_n : U \to (\mathbb{Z}/p^n\mathbb{Z})^*$. Then $U/U_1 \cong \mathbb{F}_p^*$. U_n form a decreasing sequence of open subgroups of U and $U = \varprojlim U/U_n$.

Theorem 1.13. *Suppose* $p \neq 2$ *and let* $x = p^n u$ *be an element of* $\mathbb{Q}_p^*$*, with* $u \in U$ *and* $n \in \mathbb{Z}$*. Then* x *is a square* $\iff$ n *is even and the image* $\overline{u}$ *of* u *in* $\mathbb{F}_p^* = U/U_1$ *is a square.*

Corollary 1.14. *If* $p \neq 2$*,* $\mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$ *is a group of type* $(2, 2)$*. It has for representatives* $\{1, p, u, up\}$ *where* $u \in U$ *is such that* $\left(\frac{u}{p}\right) = -1$.

Theorem 1.15. $x = p^n u \in \mathbb{Q}_2^*$ *to be a square* $\iff$ n *is even and* $u \equiv 1 \pmod{8}$.

2 Local Properties

2.1 Definition and First Properties

k denotes either $\mathbb{R}$ or $\mathbb{Q}_p$ for some prime p.

Definition 2.1. For $a, b \in k^*$, the **Hilbert symbol** (a, b) is defined as

$$(a,b) = \begin{cases} 1 & \text{if } z^2 = ax^2 + by^2 \text{ has a nontrivial solution in } k^3, \\ -1 & \text{otherwise.} \end{cases}$$

(a, b) does not change when a and b are multiplied by squares, so it defines a map $k^*/k^{*2} \times k^*/k^{*2} \to \{\pm 1\}$.

Proposition 2.2. *Let $k_b = k(\sqrt{b})$. Then $(a, b) = 1 \iff a \in Nk_b^*$, the group of norms of elements of k_b^*.*

Proof. If $b = c^2$, then $(z, x, y) = (c, 0, 1)$ is a solution. And the proposition holds since $k_b = k$.

Otherwise, β denotes a square root of b, then every element $\xi \in k_b$ can be written $m + \beta n$ with $m, n \in k$ and the norm $N(\xi) = m^2 - bn^2$.

If $a = z^2 - by^2 \in Nk_b^*$, then $(z, x, y) = (z, 1, y)$ is a solution. Coversely, if $(a, b) = 1$, then $x \neq 0$ since b is not a square. Then $a = \left(\frac{z}{x}\right)^2 - b\left(\frac{y}{x}\right)^2 = N\left(\frac{z}{x} + \beta\frac{y}{x}\right)$. $\square$

Proposition 2.3. *The Hilbert symbol satisfies:*

1. $(a, b) = (b, a)$ *and* $(a, c^2) = 1$*;*
2. $(a, -a) = 1$ *and* $(a, 1 - a) = 1$*;*
3. $(a, b) = 1 \implies (aa', b) = (a', b)$*. Actually, we will show* $(aa', b) = (a, b)(a', b)$ *later.*
4. $(a, b) = (a, -ab) = (a, (1 - a)b)$*.*

(Suppose $a \neq 1$ when the formula contains $1 - a$.)

Proof. For (3), $(a, b) = 1 \implies a \in Nk_b^* \implies (a' \in Nk_b^* \iff aa' \in Nk_b^*)$.

For (4), $(a, -a) = 1 \implies (a, b) = (a, -ab)$. $(a, 1 - a) = 1 \implies (a, b) = (a, (1 - a)b)$. $\square$

2.2 Computation of (a, b)

Lemma 2.4. *Let $v \in U$ be a p-adic unit. If $z^2 - px^2 - vy^2 = 0$ has a nontrivial solution in $\mathbb{Q}_p$, it has a solution such that $z, y \in U$ and $x \in \mathbb{Z}_p$.*

Proof. By Proposition 1.10, there exists a primitive solution (z, x, y) in $\mathbb{Z}_p$. If it did not have the desired property, then $y \equiv 0 \pmod{p}$ or $z \equiv 0 \pmod{p}$. Since $z^2 - vy^2 \equiv 0 \pmod{p}$ and $v \not\equiv 0 \pmod{p}$, we have $z \equiv y \equiv 0 \pmod{p}$, hence $px^2 \equiv 0 \pmod{p^2} \implies x \equiv 0 \pmod{p}$, contradicting the primitivity of the solution. $\square$

Theorem 2.5. *If $k = \mathbb{R}$, then $(a, b) = 1 \iff a$ or $b > 0$, and $(a, b) = -1$ otherwise.*

If $k = \mathbb{Q}_p$ and $a = p^\alpha u, b = p^\beta v$ with $u, v \in U$, the group of p-adic units, then

$$(a, b) = (-1)^{\alpha\beta\varepsilon(p)} \left(\frac{u}{p}\right)^\beta \left(\frac{v}{p}\right)^\alpha, \quad \text{if } p \neq 2$$
$$(a, b) = (-1)^{\varepsilon(u)\varepsilon(v) + \alpha\omega(v) + \beta\omega(u)}, \quad \text{if } p = 2$$

Proof. $k = \mathbb{R}$ is easy. Suppose now that $k = \mathbb{Q}_p$.

Suppose $p \neq 2$. There are only three cases of α, β to consider:

- $\alpha = \beta = 0$. It suffices to show that $(u, v) = 1$. By Corollary 1.3, $z^2 - ux^2 - vy^2 = 0$ has a nontrivial solution modulo p. Since the discriminant is a p-adic unit, by Corollary 1.12, the above solution lifts to a p-adic solution. Hence $(u, v) = 1$.

- $\alpha = 1, \beta = 0$. It suffices to show that $(pu, v) = (p, v) = \left(\frac{v}{p}\right)$. By Theorem 1.13, if v is a square, $(p, v) = \left(\frac{v}{p}\right) = 1$. Otherwise, $\left(\frac{v}{p}\right) = -1$.

- $\alpha = \beta = 1$. It suffices to show that $(pu, pv) = (pu, -p^2uv) = (pu, -uv) = \left(\frac{-uv}{p}\right) = (-1)^{\frac{p-1}{2}} \left(\frac{u}{p}\right)\left(\frac{v}{p}\right) \iff \left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$.

$p = 2$ similarly.

- $\alpha = \beta = 0$. It suffices to show that $(u, v) = 1$ if u or v is congruent to $1 \pmod 4$ and $(u, v) = -1$ otherwise.

  Suppose first that $u \equiv 1 \pmod 4$. Then by Theorem 1.15, if $u \equiv 1 \pmod 8$, u is a square and $(u, v) = 1$. Otherwise, $u + 4v \equiv 1 \pmod 8$ and then there exists $w \in U$ such that $w^2 = u + 4v$. Hence $(u, v) = 1$.

  Now suppose $u \equiv v \equiv -1 \pmod 4$. If (z, x, y) is a primitive solution of $z^2 - ux^2 - vy^2 = 0$, then $z^2 + x^2 + y^2 \equiv 0 \pmod 4 \implies x \equiv y \equiv z \equiv 0 \pmod 2$, which contradicts the primitivity of the solution. Hence $(u, v) = -1$.

- $\alpha = 1, \beta = 0$. It suffices to show that $(2u, v) = (-1)^{\varepsilon(u)\varepsilon(v)+\omega(v)}$. First we show that $(2, v) = (-1)^{\omega(v)}$, i.e. $(2, v) = 1 \iff v \equiv \pm 1 \pmod 8$. By Lemma 2.4, if $(2, v) = 1$ there exists $x, y, z \in \mathbb{Z}_2$ such that $z^2 - 2x^2 - vy^2 = 0$ and $y, z \not\equiv 0 \pmod 2$. Then $y^2 \equiv z^2 \equiv 1 \pmod 8 \implies 1 - 2x^2 - v \equiv 0 \pmod 8$. Since the only square modulo 8 are $0, 1, 4$, we have $v \equiv \pm 1 \pmod 8$.

  Conversely, if $v \equiv 1 \pmod 8$, v is a square and $(2, v) = 1$. If $v \equiv -1 \pmod 8$, $z^2 - 2x^2 - vy^2 = 0$ has $(z, x, y) = (1, 1, 1)$ for a solution modulo 8 and by Corollary 1.12, it lifts to a true solution. Hence $(2, v) = 1$.

  Now we show that $(2u, v) = (2, v)(u, v)$. It's true if $(2, v) = 1$ or $(u, v) = 1$. Hence suppose that $(2, v) = (u, v) = -1$, i.e. $v \equiv 3 \pmod 8$ and $u \equiv 3$ or $-1 \pmod 8$. Since $\frac{v}{3}$ and $\frac{u}{3}$ or $-u$ are squares, we suppose that $u = -1, v = 3$ or $u = 3, v = -5$. Now $z^2 + 2x^2 - 3y^2 = 0$ and $z^2 - 6x^2 + 5y^2 = 0$ have solution $(1, 1, 1)$, hence $(2, v) = 1$.

- $\alpha = \beta = 1$. It suffices to show that $(2u, 2v) = (2u, -4uv) = (2u, -uv) = (-1)^{\varepsilon(u)\varepsilon(-uv)+\omega(-uv)} = (-1)^{\varepsilon(u)\varepsilon(v)+\omega(u)+\omega(v)}$. Since $\varepsilon(-1) = 1, \omega(-1) = 0$ and $\varepsilon(u)(1 + \varepsilon(u)) = 0$, this is true.

$\square$

Theorem 2.6. *The Hilbert symbol is a nondegenerate bilinear form on the $\mathbb{F}_2$-vector space k^*/k^{*2}.*

Proof. If $k = \mathbb{R}$, k^*/k^{*2} is a vector space of dimension 1 (over $\mathbb{F}_2$) having $\{\pm 1\}$ for representatives.

If $k = \mathbb{Q}_p, p \neq 2$, for any $a \in k^*/k^{*2}, a \neq 1$, we find b such that $(a, b) = -1$. By Corollary 1.14, we can take $a = p, u$ or up with $u \in U$ such that $\left(\frac{u}{p}\right) = -1$, then we choose $b = u, p$ and up respectively.

If $k = \mathbb{Q}_2$, check on the representatives $\{u, 2u\}$ with $u = \pm 1, \pm 5$. Indeed we have $(5, 2u) = (-1, -1) = (-1, -5) = -1$. ☐

Remark: Write $(a, b) = (-1)^{[a,b]}$ with $[a, b] \in \mathbb{Z}/2\mathbb{Z}$. Then $[a, b]$ is a symmetric bilinear form on the k^*/k^{*2}. Theorem 2.5 gives its matrix:

- For $k = \mathbb{R}$, it is the matrix (1).
- For $k = \mathbb{Q}_p, p \neq 2$, with basis p, u where $\left(\frac{u}{p}\right) = -1$, it is the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ if $p \equiv 1 \pmod 4$ and $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ if $p \equiv -1 \pmod 4$.
- For $k = \mathbb{Q}_2$, with basis $\{2, -1, 5\}$, it is the matrix $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.

Corollary 2.7. *If b is not a square, Nk_b^* is a subgroup of index 2 in k^*.*

Proof. By Proposition 2.2, $\phi_b : k^* \to \{\pm 1\}$ defined by $\phi_b(a) = (a, b)$ has kernel Nk_b^*. Since (a, b) is nondegenerate, ϕ_b is surjective. Hence ϕ_b defines an isomorphism $k^*/Nk_b^* \cong \{\pm 1\}$. ☐

Remark: By local class field theory, let L be a finite Galois extension of k and whose Galois group G is commutative. Then $k^*/Nk^*L \cong G$ and NL^* determines L.

3 Global Properties

$(a,b)_p$ denotes the Hilbert symbol of their images in $\mathbb{Q}_p$. Define V to be the set of prime numbers together with ∞.

3.1 Product Formula

Theorem 3.1 (Hilbert). *If $a,b \in \mathbb{Q}^*$, we have $(a,b)_v = 1$ for almost all $v \in V$ and*

$$\prod_{v\in V}(a,b)_v = 1.$$

Proof. Since the Hilbert symbols are bilinear, it suffices to prove the theorem when a or b are equal to -1 or to a prime number. We use Theorem 2.5 to compute.

- $a = b = -1$. $(-1,-1)_\infty = (-1,-1)_2 = -1$ and $(-1,-1)_p = 1$ if $p \neq 2$.
- $a = -1, b = l$. If $l = 2$, $(-1,2)_v = 1$ for all $v \in V$. If $l \neq 2$, $(-1,l)_v = 1$ if $v \neq 2, l$ and $(-1,l)_2 = (-1,l)_l = (-1)^{\varepsilon(l)}$.
- $a = l, b = l'$. If $l = l'$. $(l,l)_v = (-1,l)_v$.
  If $l \neq l' = 2$, $(l,2)_v = 1$ if $v \neq 2, l$ and $(l,2)_2 = (-1)^{\omega(l)}, (l,2)_l = \left(\frac{2}{l}\right) = (-1)^{\omega(l)}$.
  Otherwise, $(l,l')_v = 1$ if $v \neq 2, l, l'$ and
  $$(l,l')_2 = (-1)^{\varepsilon(l)\varepsilon(l')}, \quad (l,l')_l = \left(\frac{l'}{l}\right), \quad (l,l')_{l'} = \left(\frac{l}{l'}\right).$$
  By Gauss quadratic reciprocity law, $\left(\frac{l'}{l}\right)\left(\frac{l}{l'}\right) = (-1)^{\varepsilon(l)\varepsilon(l')}$.

□

3.2 Existence with Given Hilbert Symbols

Lemma 3.2 (Chinese Remainder Theorem). *Let $a_1, \cdots, a_n, m_1, \cdots, m_n$ be integers with m_i being pairwise relatively prime. There exists an integer a such that $a \equiv a_i \pmod{m_i}$ for all i. Moreover, if a' is another integer with the same property, then $a' \equiv a \mod m_1 m_2 \cdots m_n$.*

Lemma 3.3 (Approximation Theorem). *Let S be a finite subset of V. The image of $\mathbb{Q}$ in $\prod_{v\in S}\mathbb{Q}_v$ is dense.*

Proof. Suppose that $S = \{\infty, p_1, \cdots, p_n\}$ and let $(x_\infty, x_1, \cdots, x_n)$ be a point of $\prod_{v\in S}\mathbb{Q}_v$. After multiplying by some integer, we may suppose $x_i \in \mathbb{Z}_{p_i}$ for $1 \leqslant i \leqslant n$.

For all integers $N > 0$, by the Chinese remainder theorem applied to $m_i = p_i^N$, there exists $x_0 \in \mathbb{Z}$ such that $v_{p_i}(x_0 - x_i) \geqslant N$ for all i.

Choose an integer $q \geqslant 2$ which is prime to all the p_i. The rational numbers of the form $\frac{a}{q^m}, a \in \mathbb{Z}, m \geqslant 0$ are dense in $\mathbb{R}$. For every $\varepsilon > 0$, choose $u = \frac{a}{q^m}$ with

$$|x_0 - x_\infty + up_1^N \cdots p_n^N| \leqslant \varepsilon.$$

Then $x = x_0 + up_1^N \cdots p_n^N$ has the property

$$|x - x_\infty| \leqslant \varepsilon, \quad v_{p_i}(x - x_i) \geqslant N.$$

□

Lemma 3.4 (Dirichlet Theorem)**.** *If a and m are relatively prime positive integers, there are infinitely many primes p such that $p \equiv a \pmod{m}$.*

Theorem 3.5. *Let $(a_i)_{i\in I}$ be a finite family of elements in $\mathbb{Q}^*$, and let $(\varepsilon_{i,v})_{i\in I, v\in V}$ be a family of numbers in $\{\pm 1\}$. Then there exists $x \in \mathbb{Q}^*$ such that $(a_i, x)_v = \varepsilon_{i,v}$ for all $i \in I$ and $v \in V \iff$ the following conditions are satisfied:*

1. *Almost all $\varepsilon_{i,v}$ are equal to 1;*
2. *For all $i \in I$, $\prod_{v\in V} \varepsilon_{i,v} = 1$;*
3. *For all $v \in V$, there exists $x_v \in \mathbb{Q}_v^*$ such that $(a_i, x_v)_v = \varepsilon_{i,v}$ for all $i \in I$.*

Proof. The necessity is trivial. Let $(\varepsilon_{i,v})$ satisfy the three conditions. After multiplying a_i by the square of some integer, we can suppose that all the a_i are integers. Let $S = \{\infty, 2\} \cup \{\text{ prime factors of } a_i\} \subseteq V$ and $T = \{v \in V | \exists i \in I, \varepsilon_{i,v} = -1\}$.

First suppose that $S \cap T = \varnothing$. Let

$$a = \prod_{l\in T} l, \quad m = 8 \prod_{l\in S, l\neq 2,\infty} l.$$

Since $S \cap T = \varnothing$, a and m are relatively prime. By Dirichlet theorem, there exists a prime $p \notin S \cup T$ such that $p \equiv a \pmod{m}$. We show that $x = ap$ has the desired property.

If $v \in S$, $\varepsilon_{i,v} = 1$, and it suffices to show that $(a_i, x)_v = 1$. $x > 0 \Longrightarrow (a_i, x)_\infty = 1$. If $v = l$ a prime number, $x \equiv a^2 \pmod{m} \Longrightarrow x \equiv a^2 \pmod{8}$ and $x \equiv a^2 \pmod{l}$ for $l \neq 2$. Since a and x are l-adic units, x is a square in $\mathbb{Q}_l^*$, hence $(a_i, x)_l = 1$.

If $v = l \notin S$, a_i is an l-adic unit, and since $l \neq 2$,

$$(a_i, b)_l = \left(\frac{a_i}{l}\right)^{v_l(b)} \quad \text{for all } b \in \mathbb{Q}_l^*.$$

If $l \notin T \cup \{p\}$, x is an l-adic unit and $v_l(x) = 0 \Longrightarrow (a_i, x)_l = 1 = \varepsilon_{i,l}$. If $l \in T$, $v_l(x) = 1$. (3) shows that there exists $x_l \in \mathbb{Q}_l^*$ such that $(a_i, x_l)_l = \varepsilon_{i,l}$. Since one of $\varepsilon_{i,l} = -1$, $v_l(x_l) \equiv 1 \pmod{2}$, hence

$$(a_i, x)_l = \left(\frac{a_i}{l}\right) = (a_i, x_l)_l = \varepsilon_{i,l}.$$

If $l = p$,

$$(a_i, x)_p = \prod_{v\neq p} (a_i, x)_v = \prod_{v\neq p} \varepsilon_{i,v} = \varepsilon_{i,p}.$$

The squares of $\mathbb{Q}_v^*$ form an open subgroup of $\mathbb{Q}_v^*$. By Approximation Theorem, there exists $x' \in \mathbb{Q}^*$ such that x'/x_v is a square in $\mathbb{Q}_v^*$ for $v \in S$. In particular $(a_i, x')_v = (a_i, x_v)_v = \varepsilon_{i,v}$ for $v \in S$. Let $\eta_{i,v} = \varepsilon_{i,v}(a_i, x')_v$, then $(\eta_{i,v})$ satisfies (1), (2), (3) and moreover $\eta_{i,v} = 1$ for $v \in S$. By the first part of the proof, there exists $y \in \mathbb{Q}^*$ such that $(a_i, y)_v = \eta_{i,v}$ and we take $x = x'y$. □

4 Quaternion Algebras

One simple presentation of $\mathbb{H}$ is

$$\mathbb{H} = \mathbb{R}\langle i, j\rangle/(i^2 + 1, j^2 + 1, ij + ji).$$

Definition 4.1. For any field k and $a, b \in k^*$, the **quaternion algebra** is

$$D_{k,a,b} = k\langle i, j\rangle/(i^2 - a, j^2 - b, ij + ji).$$

Lemma 4.2. *If $a = c^2 \in k^*$ is a square, then for any $b \in k^*$, $D_{k,a,b} \cong M_2(k)$.*

Proof. $\varphi : D_{k,a,b} \to M_2(k)$ defined by

$$\varphi(i) = \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix}, \quad \varphi(j) = \begin{pmatrix} 0 & b \\ 1 & 0 \end{pmatrix}$$

is an isomorphism. □

Proposition 4.3. *$D_{k,a,b}$ for $b \in k^*$ is isomorphic to $M_2(k)$ $\iff$ b is a norm of from $k(\sqrt{a})$, i.e.*

$$b = N((x + \sqrt{a}y)) = x^2 - ay^2, \quad \text{for some } x, y \in k.$$

Proof. It suffices to show when a is not a square. If $b = x^2 - ay^2$, then $\varphi : D_{k,a,b} \to M_2(k)$ defined by

$$\varphi(i) = \begin{pmatrix} 0 & a \\ 1 & 0 \end{pmatrix}, \quad \varphi(j) = \begin{pmatrix} x & ay \\ -y & -x \end{pmatrix},$$

gives an isomorphism.

Conversely, assume that there is an isomorphism $\varphi : D_{k,a,b} \to M_2(k)$. Since $\varphi(i)^2 = a$, there exists $g \in \mathrm{GL}_2(k)$ such that

$$g\varphi(i)g^{-1} = \begin{pmatrix} 0 & a \\ 1 & 0 \end{pmatrix}.$$

Consider a new k-algebra isomorphism $\varphi' : D_{k,a,b} \to M_2(k)$ defined by $\varphi'(q) = g\varphi(q)g^{-1}$. We put

$$\varphi'(j) = \begin{pmatrix} x & y \\ z & w \end{pmatrix} \Longrightarrow y = -az, \quad x = -w, \quad b = w^2 - az^2 \in Nk(\sqrt{a}).$$

□

Hence we have a equivalent definition of the Hilbert symbol

Definition 4.4. For $a, b \in k^*$, the Hilbert symbol (a, b) is defined as

$$(a, b) = \begin{cases} 1 & D_{k,a,b} \cong M_2(k), \\ -1 & \text{otherwise}. \end{cases}$$

Proposition 4.5. *All quaternion algebras $D_{k,a,b}$ with $(a, b) = -1$ are division rings and are isomorphic to each other. So the isomorphism classes of quaternion algebras over k is uniquely determined by the Hilbert symbol.*